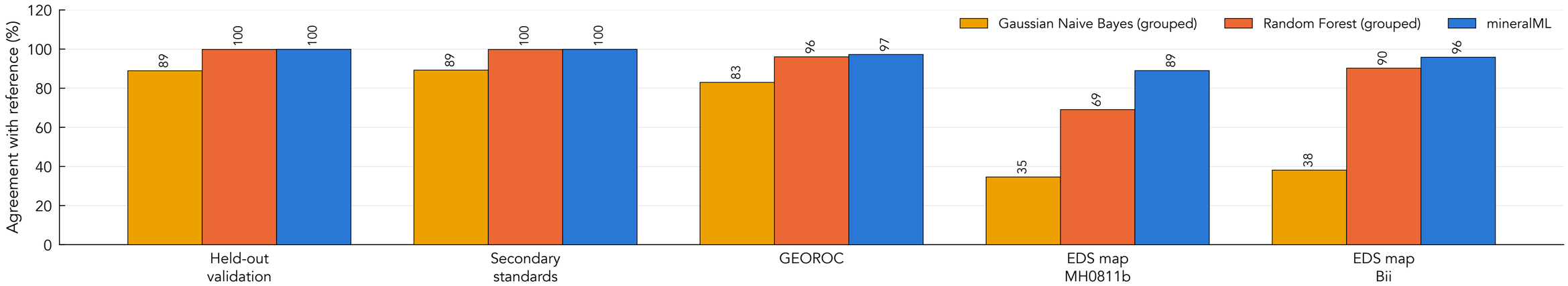
A.

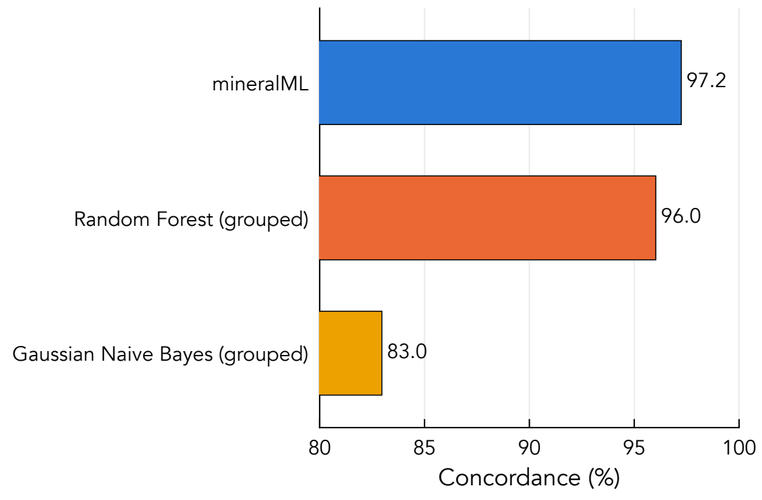


B.

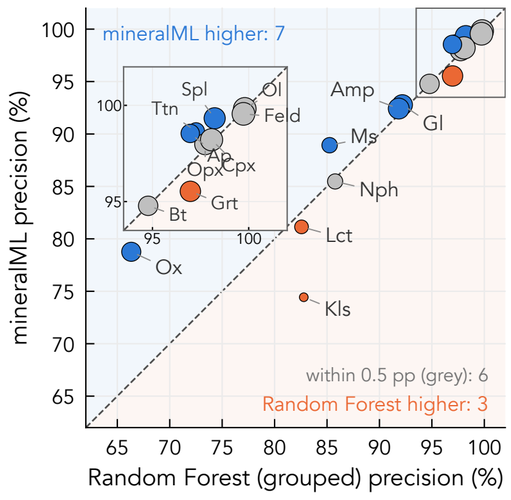
resolved by empirical composition filter

	Amphibole	Apatite	Biotite	Carbonate	Clinopyroxene	Feldspar	Garnet	Glass	Kalsilite	Leucite	Muscovite	Nepheline	Olivine	Orthopyroxene	Rhomb. Oxides	SiO_2 Polymorph	Spinel Group	Titanite	Zircon
mineralML	94	98	96	88	98	99	97	92	84	88	94	91	99	96	86	87	96	98	98
Random Forest (grouped)	94	97	96	88	98	99	98	92	87	88	92	91	97	96	76	87	94	98	98
Gaussian Naive Bayes (grouped)	81	99	92	88	96	95	93	76	28	84	93	93	70	90	65	87	90	97	98

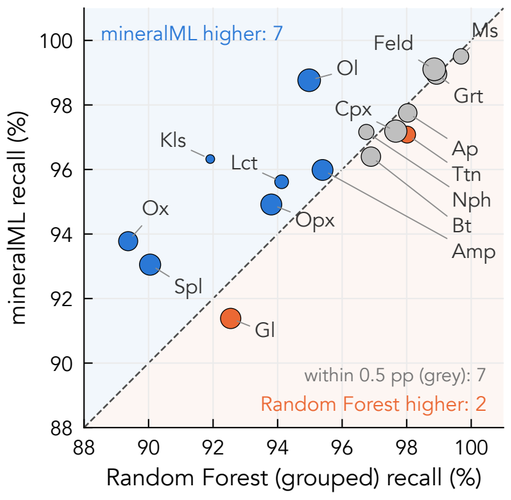
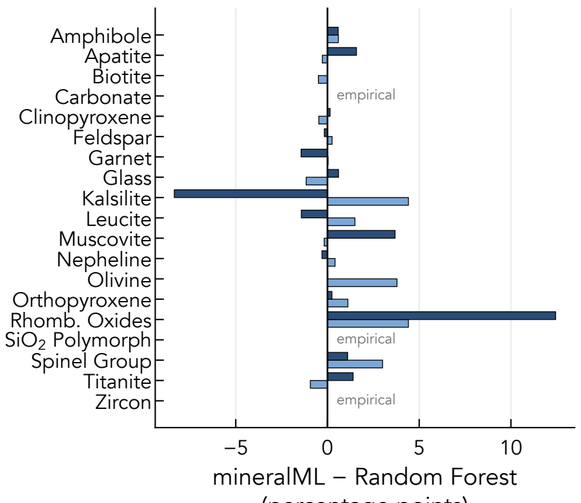
C.



D.



E.



F.